Precipitation Reactions

Sometimes it is necessary for chemists to identify the presence of ions in solution for example testing water samples. When identifying unknown positive ions and negative ions we can use **precipitation** reactions in order to determine what ions are in solution. When an ionic solid dissolves in water we call this an aqueous solution.

Some ions in aqueous solutions **react** to form **insoluble precipitates**. This is due to the **chemical properties** of the new **ionic compound** which affect its solubility in water.

Some **salts** produced from these reactions give bright coloured precipitates that are characteristic of those ions. These solids form due to strong **ionic bonds** between positively and negatively charged ions that make the salt insoluble in water.

A chemical reaction can be thought of in terms of **reactants** and **products**.
We add reactants together to get products:
REACTANTS ➔ PRODUCTS

To understand **what happens in a chemical reaction** between two ionic solutions, all you need to recognise is that the **ions simply swap partners**.

For example:

a b **c d** **a d** **c b**

Lead Nitrate + Potassium Iodide ➔ Lead Iodide (solid) + Potassium Nitrate

This is true for all precipitation reactions.

Materials Required

Dropper bottles containing solutions of:
Silver Nitrate, Barium Chloride, Sodium Hydroxide,
Sodium Chloride, Potassium Bromide, Potassium Iodide,
Copper Nitrate, Iron(II) Sulfide, Iron (III) Chloride,
Lead Nitrate and Zinc Sulfate.
8 test tubes
Test tube rack

What to do?

- Mix ten drops of each solution in separate test tubes.
- What compound forms the solid precipitate (look at solubility rules)?
- Record your observations for each reaction in the table provided.

Results Table

Reactants	Name of Ionic Solid Formed	Observations
$AgNO_3$ + NaCl	Silver Chloride	Eg A white precipitate forms when the two solutions are mixed. This indicates a chemical change has occurred.
$AgNO_3$ + KBr		
$AgNO_3$ + KI		
NaOH + FeS		
NaOH + $FeCl_3$		
NaOH + $CuNO_3$		
$Pb(NO_3)_2$ + KI		
$BaCl_2$ + $ZnSO_4$		

Solubility Rules

We know which compound is the solid due to the ions solubility in water (this is due to the chemical properties of these ions).

- **All** ionic compounds that **contain lithium, sodium or potassium** are **soluble** in water.
- **All** ionic compounds that contain **nitrate** ions are **soluble** in water.
- **All** metal **hydroxides** are insoluble in water (except LiOH, NaOH and KOH).
- Chloride, fluoride and iodide form precipitates with silver and lead ions.
- Barium ions form solids with sulfate ions.

What to do?

- Write word equations for each reaction.
- Make sure you indicate which compound forms the solid.
- Use the solubility rules to help you determine this.

Reactions of Silver Nitrate

Silver nitrate + Sodium Chloride ➔ _____

Silver Nitrate + Potassium Bromide ➔ _____

Silver Nitrate + Potassium Iodide ➔ _____

Reactions of Metal Ions with Sodium Hydroxide

Sodium Hydroxide + Copper (I) Nitrate ➔ _____

Sodium Hydroxide + Iron (II) Sulfide ➔ _____

Sodium Hydroxide + Iron (III) Chloride ➔ _____

Reactions of Lead and Zinc solutions

Potassium Iodide + Lead Nitrate ➔ _____

Barium Chloride + Zinc Sulfate ➔ _____

Extension

- Write the word equations using the correct chemical formula for each ionic compound.
- Can you Balance each chemical equation

- To correctly balance a chemical reaction you must make sure that the number of each type of atom on the left hand side is the same as the number of each type of atom on the right hand side.
- Mark in the phases of the reactants and products:
    - (aq) = aqueous solution
    - (s) = solid

Chemical Equation: Silver Nitrate + Sodium Chloride

_____

Chemical Equation: Silver Nitrate + Potassium Bromide

_____

Chemical Equation: Silver Nitrate + Potassium Iodide

_____

Chemical Equation: Sodium Hydroxide + Copper (I) Nitrate

_____

Chemical Equation: Sodium Hydroxide + Iron (III) Chloride

_____

Chemical Equation: Zinc Sulfate + Barium Chloride

_____

Chemical Equation: Lead Nitrate + Potassium Iodide

_____

Test: Chemical Formula and Chemical Equations

Q1 Name the following compounds:

I. NaCl: _____

II. KNO_3: _____

III. AgCl: _____

IV. $CuSO_4$: _____

V. $FeCl_3$: _____

VI. $Mg(OH)_2$: _____

Q2 Write the formula for the following ionic compounds:

I. Barium Chloride: _____

II. Iron (III) Sulfate: _____

III. Aluminium Hydroxide: _____

IV. Copper (II) Nitrate: _____

V. Sodium Phosphate: _____

VI. Ammonium Sulfate: _____

Q3 Explain what a precipitate is and provide an example of a reaction where a precipitate is formed?

_____

_____

_____

_____

_____

Q4a) Explain why ionic solutions are able to can conduct electricity? Provide an example and draw a diagram to show what is happening.

_____

_____

_____

_____

_____

_____

b) Is this a physical or chemical property of ions? _____

Q5 Complete the word equations for the following ionic reactions:

I. Silver Nitrate + _____ Chloride ➔ Silver_____(s) + Sodium _____
II. _____ Chloride + Zinc _____ ➔ Barium _____(s) + _____ Chloride
III. _____ Chloride + _____ Hydroxide ➔ Aluminium _____(s) + Sodium _____

Q6 Balance the following chemical equations.

I. ____H_2S + ____O_2 ➔ ____H_2O + _____SO_2
II. ____CH_4 + ____O_2 ➔ _____CO_2 + ____H_2O
III. ____$Al_2(CO_3)_3$ + ____HCl ➔ ____$AlCl_3$ + ____CO_2 + ____H_2O

Q7 Name all of the compounds in reactions I and iii (name H_2O for reaction I only):

i. _____, _____, _____, _____

iii _____, _____, _____, _____